

Satya Sai Vasamsetty

+91 93907 21741 · satya.hyd11@gmail.com · [linkedin.com/in/satya-sai-12573b1b3](https://www.linkedin.com/in/satya-sai-12573b1b3) · satzil.github.io/Portfolio/ · India

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Bash Scripting

Databases: Mysql, MongoDB, Redis, Vector DB

Concepts: Generative AI, Machine Learning, LLM, VLM, Computer Vision, Prompt Engineering, Agentic AI, RAG, Git, AWS, Linux, Edge Computing

EXPERIENCE

Senior Deep Learning Engineer

January 2024 to Present

Tarsyer Insights LLP

Pune, Maharashtra, India

- Worked on Edge Computing, Computer Vision and Gen AI Technologies and converted 9 POCs into successful production level projects spanning across 1300+ locations and increasing the revenue of the company.
- Managed 1000+ edge devices (Raspberry Pi) and created data pipelines and built automation scripts.
- Closely Interact with the clients discuss about the business requirements and provide them with suitable computer vision and Gen AI solutions. Worked with clients from Retail, Manufacturing, Medical, Logistics sectors.
- Manage and train Junior Developers, Interns potentially making them fit to handle production level projects.
- **Retail Production Monitoring System (Trent, Royaloak):** Developed a real time monitoring system for retail stores. Built 20+ use cases using computer vision and Gen AI technologies which tracks footfall, Queue Management, Intrusion, Individual person tracking, Safety and operations monitoring.
Expanded the projects to 1000+ locations for Trent, 20 locations for StarBazaar, 100+ locations for Royaloak.
Worked on Qwen and Gemini VLM, Object detection and tracking, Person Reidentification, Facial Recognition, Edge computing concepts to complete the usecases.
- **Safe Ladle Hooking system (ArcelorMittal):** Developed a Safe Ladle Hooking system which determines the hook state and triggers realtime safe and unsafe signals to the crane operators which avoids crane accidents while carrying molten metal. Deployed in 17 stations. Managed 21 cameras, 21 devices and 7 moxa devices.
Used open source object detection techniques like YOLOX , deployed onnx models for run time efficiency. Achieved 99% accuracy on all the corner cases.
- **Tool Counting and Measurement system (Oerlikon):** Developed a system where it can measure the count of tools from a metallic tool tray with 99.98% accuracy for count of tools ranging from 10 to 600.
It will also calculate the length and diameter of the tool with an error less than 0.5 mm. The setup is deployed in 2 locations significantly reducing the manual work by 90% and increased the productivity.
Used custom acrylic setup, Integrated Pi camera, Image processing, Multivariable linear regression, Object detection.
- **Warehouse and Dock monitoring (Mahindra Logistics):** Provided computer vision solutions to the logistics department and worked on 10+ usecases where it tracks dock operations and productivity, detects number plate, vehicle entry and exit time, suspicious activity, loitering time, mobile phone usage during work.
Deployed the system in 3 locations covering upto 10 docks. Used Object detection and tracking, Gemini multimodel, YoloWorld + clip.
- **Activity Assessment System (Maruti Suzuki):** Tracks the activity of the trainers in automobile plant and checks how well the activities are performed sequentially during their training process.
Monitors 10 activities and generates performance report for every trainer eliminating the need of monitoring the activities manually.
Deployed in 2 stations with integrated pi camera setup. Used image processing techniques like color segmentation, contour detection, dilation , erosion, image thresholding.
- **Patient and Operations monitoring (DrBatra's):** Developed 12 usecases which tracks productivity of the staff, clinic compliance and monitors for the patient's satisfaction in Homeopathy clinics.
Deployed in 10 clinics. Used Voice recognition, Reidentification, Image processing and classification, Object detection and tracking, Qwen VLM.

EDUCATION

Jawaharlal Nehru Technological University

Sangareddy, Telangana, India

Bachelor of Technology in Computer Science

Aug 2019 to July 2023

ACTIVITIES & ACHIEVEMENTS

- Volunteered as Python and Web Development mentor at College Technical Fest.
- Conducted an interactive Git & GitHub workshop for students at College Technical Fest.
- CODE ADDICTS: Created and hosted a programming contest at YUGA2K22 on HackerRank.